CME SPAN MARGIN MODEL

A Complete Beginner's Guide

From Zero to Portfolio Margin — Every Term, Formula & Example Explained

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1. What is Margin? (And Why Does It Exist?)

When you trade futures or options on an exchange, you don't pay the full value of the contract upfront. Instead, the exchange asks you to deposit a smaller amount of money — called margin — as a good-faith deposit to cover potential losses.

> **Think of margin like a security deposit when renting a flat. You don't pay the full rent for 12 months upfront — you just pay 1-2 months as a deposit. If you damage the property (i.e., lose money on a trade), the landlord (the exchange) uses your deposit to cover it.**

Why does margin matter?

- Exchanges need to ensure every trader can cover their losses, even if the market moves sharply overnight.
- Without margin requirements, one large trader going bust could destabilize the entire market.
- SPAN calculates exactly how much margin is needed — no more, no less — based on the actual risk of your specific portfolio.

Two types of margin

Term	Meaning
Initial Margin	The amount you must deposit before opening a trade. This is what SPAN calculates.
Maintenance Margin	The minimum balance you must keep. If your account falls below this, you get a margin call — the exchange demands you top up your account.

2. What is CME SPAN?

SPAN stands for **Standard Portfolio Analysis of Risk**. It is a margin calculation methodology developed by the **Chicago Mercantile Exchange (CME)** in 1988. It is now used by over 50 exchanges worldwide — including India's NSE (National Stock Exchange) which uses a version called SPAN-based margining through its NSCCL clearinghouse.

The core idea of SPAN — in plain English

Old-fashioned margin systems charged margin on each contract independently, ignoring the fact that some positions in your portfolio cancel each other out.

> **Example: If you are LONG (bought) 5 lots of Crude Oil March futures AND SHORT (sold) 3 lots of Crude Oil June futures, your real risk is much smaller than if you just held 5 long lots — because the two positions move in the same direction and partially offset each other.**

SPAN recognizes this. Instead of charging full margin on each leg separately, it:

1. Simulates 16 different market scenarios (different price moves and volatility changes)
2. Calculates your profit or loss (P&L) in each scenario
3. Identifies the worst-case loss across all scenarios — this becomes your Scan Risk
4. Gives you credits (reductions) for positions that offset each other
5. The final margin = Worst-case loss minus all credits

Why SPAN is better than flat per-contract margins

Feature	Old Flat System	SPAN
Recognizes offsets?	No	Yes
Accounts for option non-linearity?	No	Yes
Portfolio-level thinking?	No	Yes
Used by CME, NSE, LME, Eurex?	No	Yes

3. Key Terms & Abbreviations — Full Glossary

Every single term used in this model is explained below in plain English. Read this section first — it will make everything else much easier to understand.

A. Exchange & Market Terms

Term / Abbreviation	Full Form	Simple Explanation
CME	Chicago Mercantile Exchange	The world's largest futures exchange, based in Chicago, USA. It created the SPAN system.
SPAN	Standard Portfolio Analysis of Risk	CME's method for calculating how much margin a portfolio of futures and options needs.
NSE	National Stock Exchange	India's largest stock exchange, which uses a SPAN-based margin system.
NSCCL	National Securities Clearing Corporation Ltd	The clearinghouse of NSE — the body that actually collects and manages margin money.
LME	London Metal Exchange	The world's largest metals futures exchange, also uses SPAN.
Clearinghouse	—	The central body that sits between buyer and seller in every trade, guaranteeing that both sides will be paid. It collects margin.
Futures Contract	—	An agreement to buy or sell an asset (e.g., crude oil, gold) at a fixed price on a future date.
Options Contract	—	The right (but not obligation) to buy or sell an asset at a fixed price (the strike price) before a certain date.
Lot / Contract	—	The standardized unit of trading. 1 lot of WTI Crude Oil = 1,000 barrels. You can't buy 500 barrels — only whole lots.
Settlement Price	—	The official end-of-day price published by the exchange. All margin calculations use this price.

B. Position Terms

Term	Meaning	Example
Long	You have BOUGHT the futures contract. You profit if price goes UP.	Long 5 CL Mar25 = you bought 5 lots of March Crude Oil futures.
Short	You have SOLD the futures contract. You profit if price goes DOWN.	Short 3 CL Jun25 = you sold 3 lots of June Crude Oil futures.
Net Position	Your total signed holding. Positive = net long, Negative = net short.	+5 = net long 5 lots. -3 = net short 3 lots.
Lots	Number of contracts you hold (always positive). Direction shown separately.	5 lots long, 3 lots short.

Point Value	Dollar profit/loss for a $1 move in the futures price.	CL point value = $1,000. If oil moves $1, each lot gains/loses $1,000.
Tick Size	The minimum price increment the contract can move.	CL tick = $0.01. Oil can move $79.20 to $79.21 but not $79.205.

C. SPAN Calculation Terms

Term / Abbreviation	Full Form	Simple Explanation
Scan Risk	—	The worst-case loss your position can suffer across all 16 SPAN scenarios. This is the starting point for margin.
Scan Range	Price Scan Range	The maximum price move SPAN assumes for a given product. E.g., for CL (Crude Oil), it's $4.00 per barrel. SPAN tests what happens if price moves up to this amount.
Risk Array	—	A table showing the gain or loss for each of the 16 scenarios for a single contract. Like a lookup table of 'how much do I lose if X happens?'
SOM	Short Option Minimum	A floor margin for short options. Ensures short options always have some minimum margin even if the scenario scan says the risk is tiny (which can happen for deep OTM options).
OTM	Out of the Money	An option where exercising it would not be profitable right now. A call option is OTM if the futures price is below the strike. A put is OTM if futures price is above strike.
ITM	In the Money	An option where exercising it would be profitable right now.
ATM	At the Money	An option where the futures price roughly equals the strike price.
IV	Implied Volatility	The market's expectation of future price movement, expressed as an annual percentage. Higher IV = more expensive options = more risk.
P&L	Profit and Loss	The money gained or lost on a position.
Gross Scan Risk	—	The total scan risk before any spreading credits are subtracted. This is the 'ceiling' margin before credits.

D. Spreading Terms

Term	Full Form / Meaning	Example
Intra-commodity Spread	—	Holding opposite positions in the SAME product but DIFFERENT expiry months. E.g., Long CL March AND Short CL June. These offset each other — SPAN gives a credit.

Inter-commodity Spread	—	Holding positions in DIFFERENT but CORRELATED products. E.g., Long Crude Oil AND Long Natural Gas. These are correlated, so SPAN gives a partial credit.
Calendar Spread	—	Another name for an intra-commodity spread across different expiry months.
Spread Charge	—	A small penalty the exchange levies for each calendar spread pair, to prevent traders from gaming the margin system using fake spreads.
Spread Credit	—	The margin reduction you get for holding offsetting positions.
Credit Rate	—	The percentage of the smaller leg's scan risk that gets credited back. E.g., 50% credit rate on CL/NG means 50% of the smaller scan risk is deducted from your total margin.
Residual Risk	—	The scan risk remaining after intra-commodity credits are subtracted. Used as the input for inter-commodity credit calculation.

E. Options Terms

Term / Symbol	Full Form	Simple Explanation
Call Option	—	The right to BUY the underlying futures at the strike price. Profits when futures price rises.
Put Option	—	The right to SELL the underlying futures at the strike price. Profits when futures price falls.
Strike Price (K)	—	The fixed price at which the option holder can buy/sell. E.g., a CL Call with K=82 gives the right to buy crude oil at $82.
Premium	—	The price of the option itself. What you pay (if long) or receive (if short) to enter the option position.
Delta (Δ)	—	How much the option price moves for a $1 move in the futures price. Delta of 0.35 means the option moves $0.35 for each $1 futures move. Ranges from 0 to 1 for calls, -1 to 0 for puts.
Gamma (Γ)	—	How fast delta changes as the futures price moves. High gamma = delta is unstable and the option is sensitive to large price swings.
Vega (ν)	—	How much the option price changes for a 1% change in implied volatility. High vega = option is very sensitive to vol changes.
Theta (Θ)	—	How much value the option loses each day due to time passing (time decay). Options lose value as they approach expiry.
Black-76	—	The standard mathematical model for pricing options on futures contracts. A variant of the famous Black-Scholes model adapted for futures.

Vol Shift	Volatility Shift	In SPAN scenarios, the implied volatility is shifted up or down alongside the price move, to capture vega risk.
T (Time to Expiry)	—	How long until the option expires, measured in years. 30 days = 30/365 = 0.0822 years.
F (Futures Price)	—	The current price of the underlying futures contract.

4. Module 1 — Data Layer

Module 1 is the foundation. It defines all the inputs: what contracts exist, what positions are held, and what SPAN parameters apply to each product.

4.1 Contract Specifications

Every futures contract on an exchange has a fixed specification — standard rules that never change. In our model, a ContractSpec stores:

Field	What It Means	Example (Crude Oil)
product_code	Short code for the product	CL
product_name	Full name of the commodity	WTI Crude Oil
expiry	Which delivery month this contract is for	Mar-25 (March 2025)
contract_id	Unique identifier = code + expiry	CL_Mar25
tick_size	Minimum price movement allowed	$0.01 per barrel
point_value	Dollar value of a 1-point move per lot	$1,000 (= 1000 barrels × $1)
product_group	Category for spreading purposes	ENERGY

Why does point value matter? If crude oil moves from $79.20 to $80.20 (a $1.00 move), and you hold 5 long lots: P&L = $1.00 × $1,000 × 5 = $5,000 profit.

4.2 Portfolio Positions

The portfolio in our model has 4 futures positions:

Contract	Direction	Lots	Settlement Price	Why Interesting
CL_Mar25	LONG	5	$79.20	Part of a calendar spread with CL_Jun25
CL_Jun25	SHORT	3	$80.10	Short leg of the CL calendar spread
NG_Mar25	LONG	4	$2.71	Natural Gas — correlated with CL for inter-commodity spread
GC_Apr25	LONG	2	$2,025.00	Gold — different product group, no inter-commodity credit

4.3 SPAN Parameters

Each product has its own SPAN parameters set by the exchange:

Parameter	Meaning	CL Value	NG Value	GC Value
Price Scan Range	Max 1-day price move SPAN assumes	$4.00/bbl	$0.50/MMBtu	$35.00/oz

Intra Spread Charge	Penalty per calendar spread pair	$200	$150	$100
Inter Commodity Credit Rate	% credit for correlated product pairs	50%	50%	0%

4.4 The 16-Scenario Grid

SPAN tests 16 scenarios, each defined by a price fraction and a volatility shift direction:

Scenario	Price Fraction	Vol Shift	Weight	What It Tests
S1	+1/3	Up	1.00	Small price rise, vol up
S2	+1/3	Down	1.00	Small price rise, vol down
S3	+2/3	Up	1.00	Medium price rise, vol up
S4	+2/3	Down	1.00	Medium price rise, vol down
S5	+1x	Up	1.00	Full scan range rise, vol up
S6	+1x	Down	1.00	Full scan range rise, vol down
S7	+2x	Up	1.00	2x scan range rise, vol up (stress test)
S8	+2x	Down	1.00	2x scan range rise, vol down (stress test)
S9	-1/3	Up	1.00	Small price fall, vol up
S10	-1/3	Down	1.00	Small price fall, vol down
S11	-2/3	Up	1.00	Medium price fall, vol up
S12	-2/3	Down	1.00	Medium price fall, vol down
S13	-1x	Up	1.00	Full scan range fall, vol up
S14	-1x	Down	1.00	Full scan range fall, vol down
S15	+2x	Up	0.35	Extreme rise — WEIGHTED at 35%
S16	-2x	Down	0.35	Extreme fall — WEIGHTED at 35%

Why are S15 and S16 only weighted 35%? These extreme scenarios (price moves 2x the scan range) are possible but considered less likely than the normal scenarios. SPAN still tests them — but gives them only 35% weight, so their contribution to scan risk is proportionally smaller.

5. Module 2 — Risk Array Engine

Module 2 is the heart of SPAN. It calculates how much money each position would lose in each of the 16 scenarios.

5.1 Scenario P&L Formula (Futures)

For a futures position, the P&L in any scenario is straightforward:

```
Scenario P&L = Price Move × Point Value × Net Position
```

Where Price Move = Price Fraction × Price Scan Range

Worked Example — CL_Mar25 (Long 5 lots, settlement $79.20, scan range $4.00): Scenario 5 (price fraction = +1.0, full scan range up): Price Move = +1.0 × $4.00 = +$4.00 Hypo Price = $79.20 + $4.00 = $83.20 P&L = $4.00 × $1,000 × 5 = +$20,000 (profit — price went up, we are long) Scenario 13 (price fraction = -1.0, full scan range down): Price Move = -1.0 × $4.00 = -$4.00 Hypo Price = $79.20 - $4.00 = $75.20 P&L = -$4.00 × $1,000 × 5 = -$20,000 (loss — price went down, we are long)

5.2 Extreme Scenario Weighting

Scenarios 15 and 16 are extreme — price moves 2x the scan range. Their raw P&L is calculated first, then multiplied by 0.35:

```
Weighted P&L (S15/S16) = Raw P&L × 0.35
```

For CL_Mar25 in S16: Raw P&L = -$8.00 × $1,000 × 5 = -$40,000 → Weighted = -$40,000 × 0.35 = -$14,000

5.3 Scan Risk Definition

After computing all 16 weighted P&Ls, the scan risk for a position is simply:

```
Scan Risk = max(0, -min(Weighted P&L across all 16 scenarios))
```

i.e., the largest loss across all scenarios, floored at zero (you can't have negative scan risk)

Results for our portfolio:

Contract	Position	Worst Scenario	Worst P&L	Scan Risk
CL_Mar25	Long 5 lots	S13 (price -$4.00)	-$20,000	$20,000
CL_Jun25	Short 3 lots	S7 (price +$8.00)	-$24,000	$24,000
NG_Mar25	Long 4 lots	S13 (price -$0.50)	-$20,000	$20,000
GC_Apr25	Long 2 lots	S13 (price -$35.00)	-$7,000	$7,000
TOTAL	—	—	—	$71,000

Note on CL_Jun25: It is SHORT, so it LOSES money when price goes UP. That's why its worst scenario is S7 (large price rise), not S13 (price fall). The scan always finds the worst case for the actual direction of your position.

6. Module 3 — Spreading Credits

The gross scan risk of $71,000 is the ceiling. Module 3 reduces it by recognizing positions that offset each other.

6.1 Intra-Commodity (Calendar) Spread Credit

We hold Long 5 CL_Mar25 and Short 3 CL_Jun25 — both are crude oil, just different months. When oil prices move, both contracts move in the same direction. So the long and short partially cancel each other out.

How many spread pairs are formed?

```
Number of Spreads = min(Long Lots, Short Lots) = min(5, 3) = 3
```

3 spread pairs are formed. The remaining 2 long lots are unhedged.

For each spread pair, SPAN gives credit equal to the per-lot scan risk of both legs:

```
Per-lot scan risk (CL_Mar25) = $20,000 ÷ 5 lots = $4,000 per lot
Per-lot scan risk (CL_Jun25) = $24,000 ÷ 3 lots = $8,000 per lot
```

Gross Credit = ($4,000 + $8,000) × 3 spreads = $36,000

But the exchange adds a spread charge penalty to prevent gaming:

```
Spread Charge = $200 per spread × 3 spreads = $600 Net Intra
Credit = $36,000 - $600 = $35,400
```

The $200/spread charge is a small fee to discourage traders from creating fake calendar spreads just to reduce margin.

Intuition: If you are long March oil and short June oil, and oil prices crash, you lose on March but gain on June. Your NET loss is much smaller than holding just the March position alone. SPAN rewards this with a $35,400 credit.

6.2 Inter-Commodity Spread Credit

We hold Long NG_Mar25 (Natural Gas) and Long CL_Mar25 (Crude Oil). These are different products, but they are correlated — when oil prices rise, gas prices often rise too.

The inter-commodity credit works on RESIDUAL scan risk — the scan risk that remains after intra-commodity credits have been applied:

```
CL Residual Risk = ($20,000 + $24,000) - $35,400 = $8,600 NG
Residual Risk = $20,000 (no intra credit for NG)  Inter Credit =
50% × min($8,600, $20,000) = 50% × $8,600 = $4,300
```

The credit is always based on the SMALLER of the two residual risks — the 'binding leg'.

Why min()? SPAN gives credit only up to the size of the smaller side. If you have $8,600 of CL residual risk and $20,000 of NG risk, the CL side is the binding constraint. You can't get more credit than your smallest correlated exposure.

6.3 Spreading Summary

Component	Amount	Cumulative Net Risk
Gross Scan Risk	+$71,000	$71,000
(-) Intra-Commodity Credit (CL calendar)	-$35,400	$35,600
(-) Inter-Commodity Credit (CL/NG 50%)	-$4,300	$31,300
= Net Scan Risk (pre-aggregation)	—	$31,300

The spreading credits reduced total margin by $39,700 — a 55.9% reduction. This is the power of SPAN.

7. Module 4 — Margin Aggregator

Module 4 allocates the spreading credits back to individual positions and computes the final SPAN margin per position.

7.1 Credit Allocation Logic

The total credits ($35,400 intra + $4,300 inter) need to be distributed across the four positions. The logic:

- Intra credit: Distributed proportionally across the paired legs (CL_Mar25 gets its long-leg share, CL_Jun25 gets its short-leg share), based on each leg's gross contribution to the spread credit.
- Inter credit: Goes entirely to the smaller residual risk leg (CL in this case, since $8,600 < $20,000), distributed proportionally across CL contracts by their scan risk share.
- Unhedged positions (NG_Mar25, GC_Apr25): Get no intra credit (no opposing leg). May receive reallocated excess credit.

7.2 The Credit Leakage Problem & Reallocation Fix

A technical problem arises: CL_Jun25 initially received more credit than its scan risk. Specifically, its total allocated credit exceeded its $24,000 scan risk — meaning its margin would be negative, which is impossible (margin is floored at zero).

The excess credit doesn't disappear — it gets REALLOCATED. An iterative loop redistributes the overflow to other positions that still have remaining margin capacity. This ensures the portfolio total margin exactly equals Gross Scan Risk minus total credits ($71,000 - $39,700 = $31,300).

7.3 Final Per-Position Margin

Contract	Scan Risk	Intra Credit	Inter Credit	SPAN Margin
CL_Mar25	$20,000	$11,800	$1,955	$6,245 approx
CL_Jun25	$24,000	$23,600	$2,345	$0 (fully offset)
NG_Mar25	$20,000	$0	$1,170 (reallocated)	$18,830 approx
GC_Apr25	$7,000	$0	$410 (reallocated)	$6,590 approx
TOTAL	$71,000	$35,400	$4,300	~$31,300

CL_Jun25 ends up with $0 margin. This makes economic sense — the short June position is almost entirely offset by the long March position. The NET crude oil risk is small, concentrated in the 2 unhedged long lots of March.

8. Module 5 — Output Layer

Module 5 is purely presentational — it assembles everything from Modules 1-4 into a clean, readable report. No new calculations happen here.

The report has 5 sections:

Section	Contents
Section 1	Portfolio Positions — all contracts, directions, sizes, prices
Section 2	Scenario Analysis Summary — best/worst scenario per contract, scan risk
Section 3	Spreading Credits — intra and inter commodity credit detail
Section 4	Per-Position SPAN Margin — the final margin table
Section 5	Margin Waterfall — step-by-step reduction from gross to net margin

The master function run_span_model() runs all modules in sequence and returns all computed objects as a dictionary, so you can access any intermediate result programmatically after the run.

9. Module 6 — Options Support

Module 6 extends the model to handle options on futures contracts. Options are fundamentally different from futures because their P&L is non-linear — the same price move affects an option differently depending on how far the option is from the money.

9.1 Black-76 Options Pricing Model

For pricing options on futures, we use the Black-76 model (a variant of the famous Black-Scholes model developed by Fischer Black in 1976).

The inputs to Black-76 are:

Input	Symbol	Meaning	Example (CL_Mar25_C82)
Futures Price	F	Current settlement price of the underlying futures	$79.20
Strike Price	K	The price at which the option can be exercised	$82.00
Time to Expiry	T	Years until option expires (e.g. 30 days = 30/365)	0.0822 years
Implied Volatility	σ (sigma)	Market's forecast of price variability. 28% = price moves ~28% per year.	0.28 (28%)
Option Type	—	Call (right to buy) or Put (right to sell)	Call

The Black-76 formula for a CALL option:

```
d1 = [ln(F/K) + 0.5×σ²×T] / (σ×√T)  d2 = d1 - σ×√T  Call Price = e^(-rT) × [F×N(d1) - K×N(d2)]
```

Where N(x) = cumulative normal distribution, r = risk-free rate (set to 0 for margining purposes), e = Euler's number (~2.718)

Plain English: The Black-76 formula asks — given where the futures price is today (F), what is the probability-weighted expected payoff of this option at expiry? The higher the volatility (σ) and the more time remaining (T), the more the option is worth.

9.2 Options Scenario P&L

Unlike futures (where scenario P&L is just price_move × point_value × lots), for options we must RE-PRICE the option under each scenario and compute P&L as the difference:

```
Scenario P&L = (Scenario Premium - Current Premium) × Point Value × Net Position
```

Where Scenario Premium = Black-76 price using (F + price_move) as the new futures price and (IV ± vol_shift) as the new volatility

This means each scenario has BOTH a price move AND a vol shift:

What Changes	How	Effect on Options
Futures Price	Moves by price_fraction × scan_range	Changes intrinsic value — delta effect
Implied Volatility	Shifts up (+5% for CL) or down (-5% for CL)	Changes extrinsic value — vega effect
Both together	Price + Vol change simultaneously	The worst case is usually price AND vol moving adversely together

9.3 The Greeks — What They Tell You

The Greeks quantify how sensitive an option is to various market inputs:

Greek	Symbol	Measures Sensitivity To...	Our CL Call (C82)	Interpretation
Delta	Δ	Change in futures price	0.3473	For each $1 crude oil moves, this call moves $0.3473. It's OTM, so delta < 0.5.
Gamma	Γ	Rate of change of delta	0.0581	Delta itself changes — as futures moves up, delta increases. High gamma = unstable risk profile.
Vega	v	Change in implied volatility	0.0839	For each 1% rise in IV, this call gains $0.0839 per unit. High vega = expensive in volatile markets.
Theta	Θ	Passage of time (time decay)	Negative	Short-dated options lose value each day. Being short an option actually benefits from theta.

9.4 Short Option Minimum (SOM)

The SOM is a safety floor for short option positions. Here's the problem it solves:

- Suppose you sold a DEEP OTM call — strike $150 when oil is at $79. The scenario scan says: even if oil moves $8 up to $87, this deep OTM call barely changes value. Scan risk ≈ $0.
- But there's still real tail risk — oil could spike to $120 in an extreme event not captured by the 16 scenarios.
- SOM prevents margin from going to near-zero for these positions.

```
SOM = SOM Rate × Price Scan Range × Point Value × |Lots| SOM Rate = 50% (for CL, NG, GC in our model)
```

Example: CL_Mar25_C82 (Short 3 lots): SOM = 50% × $4.00 × $1,000 × 3 = $6,000

```
Final Option Risk = max(Scan Risk from 16 scenarios, SOM)
```

SPAN takes whichever is higher — the scenario scan or the SOM floor.

In our model, the SOM did NOT bind for the CL call ($6,000 SOM vs $15,072 scan risk) because the option was not deep OTM — delta of 0.35 means it has meaningful price sensitivity.

9.5 Options Results Summary

Option	Type	Strike	Moneyness	Premium	Delta	Scan Risk	SOM	SOM Bin
CL_Mar25_C82	Short Call	$82.00	OTM	$1.4183	0.3473	$15,072	$6,000	No
CL_Mar25_P76	Long Put	$76.00	OTM	$1.0512	-0.2774	$2,065	$0	N/A (long
GC_Apr25_C2100	Short Call	$2100	OTM	$22.87	0.2933	$3,928	$1,750	No

Component	Amount
Futures SPAN Margin (from Module 4)	$31,300
Options Risk — CL_Mar25_C82 (short call)	$15,072
Options Risk — CL_Mar25_P76 (long put)	$2,065
Options Risk — GC_Apr25_C2100 (short call)	$3,928
TOTAL PORTFOLIO MARGIN (Futures + Options)	$52,365

10. End-to-End Worked Example

Let's trace a single position — CL_Mar25 (Long 5 lots of WTI Crude Oil March futures) — through every step of the model.

Step 1: Identify the position

Field	Value
Contract ID	CL_Mar25
Direction	LONG (we bought the futures)
Number of Lots	5
Settlement Price	$79.20 per barrel
Point Value	$1,000 per lot per $1 move
Price Scan Range	$4.00 per barrel

Step 2: Compute Scenario P&L for Key Scenarios

Scenario	Price Fraction	Price Move	Hypo Price	P&L Calculation	Weighted P&L
S5 (best)	+1.0x	+$4.00	$83.20	+$4.00 × $1,000 × 5	+$20,000
S13 (worst)	-1.0x	-$4.00	$75.20	-$4.00 × $1,000 × 5	-$20,000
S7 (stress up)	+2.0x	+$8.00	$87.20	+$8.00 × $1,000 × 5	+$40,000
S16 (extreme down)	-2.0x	-$8.00	$71.20	-$8.00 × $1,000 × 5 × 0.35	-$14,000

Step 3: Determine Scan Risk

```
Scan Risk = max(0, -min(all weighted P&Ls))            = max(0,
                -(-$20,000))              = $20,000
```

The worst-case weighted P&L is -$20,000 (Scenario 13). Scan risk = $20,000.

Step 4: Apply Intra-Commodity Spread Credit

CL_Mar25 is paired with CL_Jun25 (3 spread pairs). CL_Mar25 gets the long-leg credit:

```
Long-leg gross credit = $4,000/lot × 3 spreads = $12,000 Pro-
rated net intra credit allocated to CL_Mar25 ≈ $11,800 (slightly
less than $12,000 due to spread charge allocation)
```

After reallocation, CL_Mar25 receives approximately $11,800 of intra credit.

Step 5: Apply Inter-Commodity Spread Credit

CL_Mar25 is part of the CL/NG pair. CL gets the inter credit (smaller residual risk side):

```
Inter credit allocated to CL_Mar25 ≈ $1,955 (proportional share
  of the $4,300 total inter credit, based on scan risk share
                        within CL)
```

CL_Mar25's share = ($20,000 / $44,000) × $4,300 = $1,955

Step 6: Final SPAN Margin

```
SPAN Margin (CL_Mar25) = Scan Risk - Intra Credit - Inter Credit
 = $20,000 - $11,800 - $1,955                          ≈ $6,245
```

Of the original $20,000 scan risk, only ~$6,245 is the final margin required for this position.

11. How to Run the Model

11.1 Requirements

- Python 3.8 or higher
- pandas library: pip install pandas
- numpy library: pip install numpy
- scipy library: pip install scipy (required for Module 6 options only)

11.2 File Structure

All 6 modules must be pasted into a single Python file in order, or each module can be saved as a separate .py file and imported. The recommended single-file structure:

```
span_model.py
├── Module 1: Data Layer (contracts, positions, SPAN params, scenarios)
├── Module 2: Risk Array Engine (scenario P&L, scan risk)
├── Module 3: Spreading Module (intra/inter credits)
├── Module 4: Margin Aggregator (credit allocation, reallocation loop)
├── Module 5: Output Layer (report printing, run_span_model())
└── Module 6: Options Support (Black-76, SOM, run_span_model_v2())
```

11.3 Running the Futures-Only Model

```
# Paste Modules 1-5, then run:
python span_model.py

# Or in a Jupyter notebook:
results = run_span_model()

# Access specific outputs:
results['margin_df']       # Per-position margin table
results['port_summary']    # Portfolio margin summary dict
results['spreading']       # All spreading credit details
results['scan_summary']    # Scan risk per position
```

11.4 Running the Full Model with Options

```
# Paste all Modules 1-6, then run:
results = run_span_model_v2()

# This runs the complete pipeline and prints a 6-section report.
# Additional output keys:
results['combined_df']     # Futures + options scan risk combined
results['total_margin']    # Total portfolio margin (futures + options)
```

11.5 Customising the Portfolio

To change positions, edit the PORTFOLIO dictionary in Module 1:

```
# Example: Change CL_Mar25 from long 5 to long 10:
PORTFOLIO['CL_Mar25'] = Position(
    contract_id    = 'CL_Mar25',
    net_position   = +10,    # Changed from +5 to +10
    entry_price    = 78.50,
    current_price  = 79.20
)

# Then re-run:
results = run_span_model()
```

To add a new product, add entries to CONTRACT_SPECS, SPAN_PARAMS, and PORTFOLIO in Module 1. If it should have inter-commodity spread credit with an existing product, also add a row to INTERCOMMODITY_SPREADS.

12. Limitations & What Real CME SPAN Does Differently

This is a dummy model built for learning. Here is an honest comparison with the actual CME SPAN system:

Feature	Our Dummy Model	Real CME SPAN
Scenario grid	16 fixed scenarios, same for all products	16 scenarios per product, parameters set daily by CME based on actual market volatility
Scan range source	Hardcoded ($4.00 for CL, etc.)	Published daily in CME's SPAN parameter files (.pa2 format) based on 99th percentile of historical returns
Options spreading	Options add to futures margin independently	Options can also form spread relationships with futures and other options (delta-based spread credits)
Volatility surface	Single flat IV per option	Full implied volatility surface — different IV for each strike and expiry
Cross-margining	No	CME offers cross-margining across related products and even across exchanges (e.g., CME + CBOT)
Currency handling	All USD	Multi-currency with FX risk conversion
SPAN 2	Not implemented	CME launched SPAN 2 in 2022 — more granular scenario grid, improved Greeks-based decomposition
Delivery charges	Not implemented	Positions near expiry get an additional delivery/spot charge in real SPAN
Inter-commodity tiers	Single flat credit rate	CME uses a tiered priority system for inter-commodity credits — multiple spread tiers with different credit rates
Floating point precision	~$0.01 rounding residual	Production SPAN rounds to cent at each step, eliminating rounding artifacts

What this model IS good for

- Understanding the conceptual logic of SPAN from first principles
- Learning how scenario analysis, scan risk, and spreading credits work together
- Building intuition for how portfolio composition affects margin (calendar spreads, correlation credits)
- A foundation for building a more complete implementation — all the core machinery is here

— End of README —

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